

2. Human haemoglobin has a quaternary structure consisting of 2 alpha and 2 beta polypeptide chains. A mutation can result in the production of a beta chain which causes a red blood cell to become distorted into a sickle shape at low partial pressures of oxygen.

The following table shows the genotypes present in a population, where the allele for normal haemoglobin is HbA and for the abnormal haemoglobin is HbS.

Genotype	Description
HbA HbA	Haemoglobin does not cause red blood cells to sickle.
HbA HbS	Codominant alleles producing 50 % normal haemoglobin and 50 % mutated. Sickling of red blood cells can take place at low partial pressures of oxygen. This is called sickle cell trait.
HbS HbS	All haemoglobin is affected which results in severe sickling of red blood cells. Red blood cells only live for 10 – 20 days causing severe anaemia (sickle cell anaemia). Without treatment, it is rare for individuals homozygous for this condition to live to reproductive age.

- (a) Using the equations below, calculate the frequency of alleles HbA and HbS in African-Americans living in America where 1 in 625 suffers from sickle cell anaemia.

Use these values to calculate the percentage frequency of heterozygotes in the population. [4]

p = frequency of HbA

q = frequency of HbS

$p + q = 1$

$p^2 + 2pq + q^2 = 1$

frequency of HbA =

frequency of HbS =

frequency of heterozygotes =

% frequency of heterozygotes =



- (b) Malaria is the most serious and wide-spread parasitic disease in humans; the vast majority of global malaria cases are in Africa.

Sickle red blood cells are less likely to be infected by the malarial parasite.

The frequency of heterozygotes (HbA HbS) in some parts of Africa is approximately 42 %.

The frequency of heterozygotes in African-Americans is lower than in native African populations.

Using the information provided and your own knowledge, explain the differences in the percentage frequency of African-American heterozygotes compared to those in Africa.

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- (c) Malaria is caused by a parasite of the genus *Plasmodium* and it is transmitted from one person to another by *Anopheles* mosquitoes. In 2002, after seven years, the Plasmodium Genome Project and the Anopheles Genome Project were both completed.

- (i) State what information has been obtained from the Anopheles and Plasmodium Genome Projects. [2]

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- (ii) State **two** uses of the information obtained from these genome projects in the control of malaria. [2]

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